# Juan de Dios Alvarez

**Senior Data Scientist - Machine Learning, Model Evaluation and Applied AI**

gmail@juandedios.info | +52 56 5495 9618 | Morelia, Mexico | linkedin.com/in/juandedios-info

profile.juandedios.info/978822

## Summary

Data scientist with seven years of industry work (2019 to present) and a research background in physics and computer science. Builds models and pipelines in Python and SQL that a team can rerun, audit and maintain, and measures each method against a baseline, control set or ground truth before anyone relies on it, documenting its assumptions and limits. Works AI-first with Claude daily and decides between an LLM and a classical method by measurement. Learns new tools on the job and puts them to use within the engagement, as two recommendation letters describe independently.

## How I Work

- **Ramp-up on unfamiliar technology:** studied LabVIEW on arrival at CERN and the University of Lausanne and, in the same one-month stay, developed the data-acquisition communication software for a PET scanner prototype. Adopted Agile (Jira, Scrum) at D3Clarity without prior experience, per a recommendation letter.
- **Critical review of AI output:** checks LLM results against rule-based baselines, labeled control cases and source text; adopts AI-generated code or extractions only after they pass mechanical checks.
- **Communication with non-technical audiences:** turned analyses into dashboards and executive reports used directly by client sales teams and decision-makers; taught KPI dashboarding at a business school.
- **Team development:** founded and led a student Data Science group, supervised an undergraduate team at FIRA, and documented conventions so work did not depend on one person.

## Experience

### Data Scientist & ML Engineer | Independent Practice

**March 2026 - Present**

- Compared document-extraction methods (classical OCR, layout detectors, local vision and language models) head to head with character and word error rates against a ground-truth calibration set; adopted changes only on measured before/after results (doubtful lines reduced from 175 to 77 on a fixed sample).
- Evaluated LLM extraction against a regex baseline (recall, false positives, cost, runtime) and gated its use on a labeled control set of 26 positive and negative cases with an explicit acceptance criterion.
- Measured keyword, embedding and hybrid retrieval by recall@k against a hand-labeled question bank, documented the benchmark's known bias, and estimated misses with fixed-seed stratified samples.
- Built unsupervised categorization of short Spanish text (embeddings, UMAP, HDBSCAN, FINCH) scored by silhouette, DBCV and ARI, with a KL-divergence screen for clusters explained by metadata.
- Designed and run the data system of a small retail business in production: PostgreSQL warehouse (40+ tables, PL/pgSQL), FIFO inventory valuation, and bank-to-invoice reconciliation with confidence tiers.

### Economic Research Specialist (Data Scientist) | FIRA

**January 2025 - January 2026**

- Replaced one analyst's manual Excel process with a scheduled pipeline: 74 extraction scripts (Python, Playwright) over 9+ public sources (INEGI, Banxico, USDA, others) loading PostgreSQL with provenance.
- Wrote 60+ PostgreSQL functions and indexed materialized views (window functions, JSONB, CTEs) computing economic indicators: inflation contributions, trade balances, supply-and-demand balances, crop yields.
- Built a crop-production model from Sentinel-2 and Planet Labs vegetation indices: outlier filtering, PCA, Random Forest and Gradient Boosting selected by 5-fold cross-validated RMSE, with held-out RMSE, MAE and R2.

- Delivered automated executive Excel reports and 15+ Apache Superset dashboards to decision-makers.
- Set governance conventions (raw data preserved, lineage, GitLab); supervised an undergraduate team.

### Data Scientist | CITEIM

**October 2024 - January 2025**

- Scoped a Mexican Sign Language recognition system from a literature review and built its dataset: video sourcing via APIs, weak labels from subtitles (OCR, speech-to-text), MediaPipe keypoints.

### Data Scientist & Team Lead | Punto Singular

**December 2023 - August 2024**

- Led monthly analytics cycles for an EdTech client, from metric definition to results; KPI dashboards in Tableau and Looker.
- Founded and led a Data Science group of undergraduate students working on real client and nonprofit projects.

### Data Scientist | BairesDev

**September 2023 - November 2023**

- Built a retrieval-augmented generation proof of concept that let a legal client's lawyers query their contracts for obligations, risk clauses and precedents; also delivered transformer-based summarization and Neo4j graph modeling.

### Data Scientist | D3Clarity (formerly Viqtor Davis North America)

**February 2019 - March 2023**

- First data scientist of the company; defined the analytics roadmap for US-based clients in a distributed team.
- Classified school-board minutes with NLP on Azure ML; built a Power BI dashboard for the client's sales team.
- Built consumer-segment forecasting models, data lakes, automated ingestion and ETL on AWS.

### Lecturer & Mentor | Tecnologico de Monterrey, Campus Morelia

**2013 - 2014; 2023 - 2024**

- Taught KPI dashboarding (Tableau) at the Business School, plus math, physics and programming; mentored a robotics team.

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## Education

- **PhD Candidate in Physics** - University of Michoacan (2009 - 2013). HAWC data analysis; 14 publications.
- **Master's in Computer Science** - INAOE (2006 - 2009). Thesis: text classification by prototype-based class reduction.
- **Bachelor's in Physics and Mathematics** - University of Michoacan (1998 - 2003).

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## Skills

- **Python and ML:** pandas, Polars, NumPy, SciPy, scikit-learn (Random Forest, Gradient Boosting, PCA, cross-validation), sentence-transformers, HDBSCAN/FINCH/UMAP, RAG, NER (GLiNER); PyTorch at prototype level
- **Evaluation and statistics:** baselines and ground truth, control sets, k-fold cross-validation, RMSE/MAE/R2, recall@k, CER/WER, silhouette/DBCV/ARI, KL divergence, stratified sampling, documented benchmark bias
- **SQL, data and reporting:** PostgreSQL (PL/pgSQL, window functions, materialized views, JSONB), SQLite, Oracle, MySQL, Neo4j; ETL, Playwright, APIs, lineage; Superset, Power BI, Tableau, Looker; R
- **Engineering and cloud:** Git (GitHub, GitLab), modular code, locked dependencies (uv), credentials outside version control; AWS (EC2, SageMaker), Azure ML, Google Cloud service accounts
- **LLMs:** Claude Code, local LLMs (Ollama), schema-constrained extraction, source-grounding checks

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## Certifications and Languages

- AWS Certified Cloud Practitioner; Semarchy xDM
- Spanish (native), English (professional), Italian (basic-intermediate), French (basic)